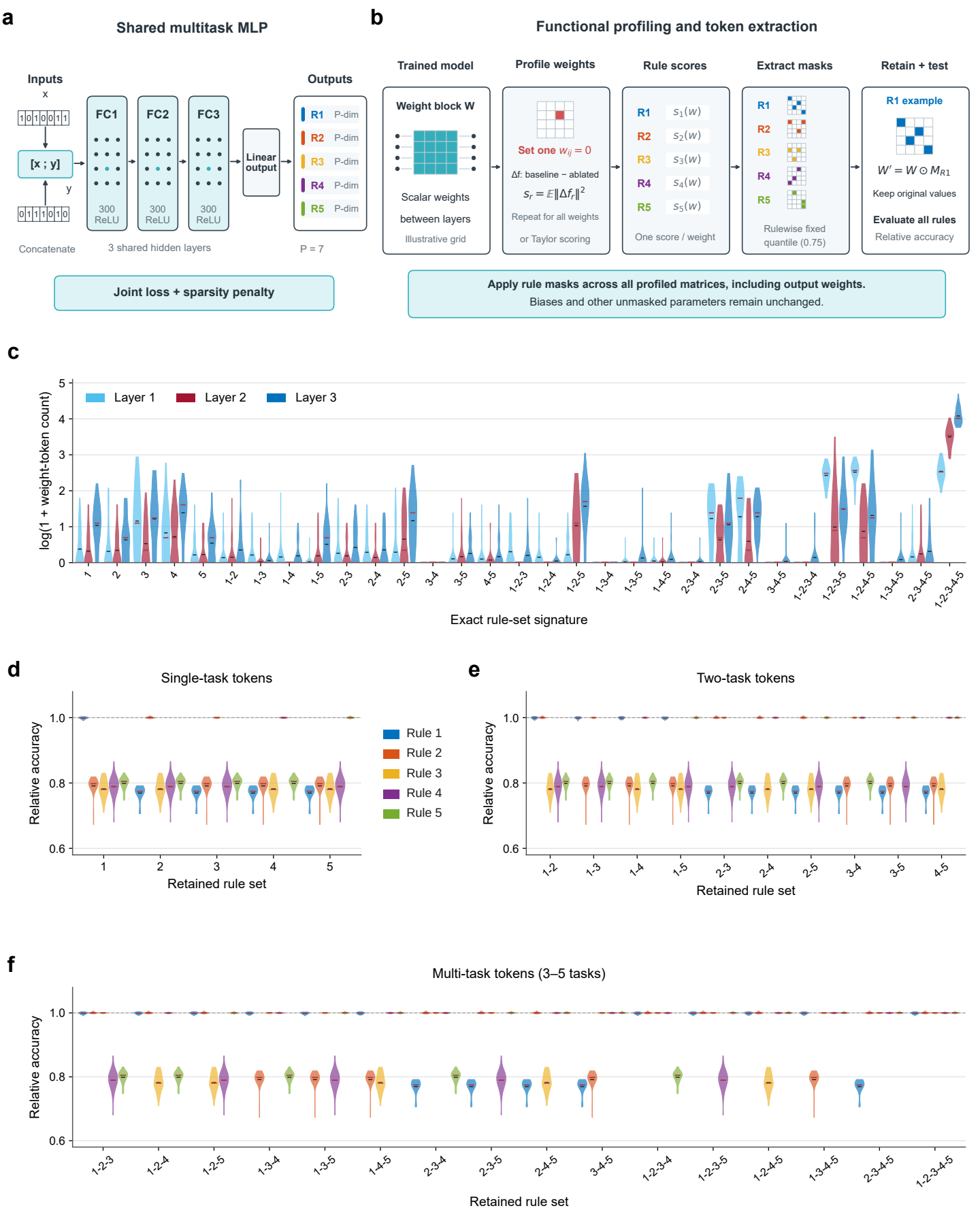

a
Shared multitask MLP
Inputs
x
y
[x ; y]
Concatenate
FC1
FC2
FC3
300 ReLU
3 shared hidden layers
Linear output
Outputs
R1 P-dim
R2 P-dim
R3 P-dim
R4 P-dim
R5 P-dim
P = 7
Joint loss + sparsity penalty
b
Functional profiling and token extraction
Trained model
Weight block W
Scalar weights between layers
Illustrative grid
Profile weights
Set one $w_{ij} = 0$
Δf: baseline − ablated
$s_r = \mathbb{E}\|\Delta f_r\|^2$
Repeat for all weights
or Taylor scoring
Rule scores
$s_1(w)$
$s_2(w)$
$s_3(w)$
$s_4(w)$
$s_5(w)$
One score / weight
Extract masks
Rulewise fixed quantile (0.75)
Retain + test
R1 example
$W' = W \odot M_{R1}$
Keep original values
Evaluate all rules
Relative accuracy
Apply rule masks across all profiled matrices, including output weights.
Biases and other unmasked parameters remain unchanged.
c
Layer 1
Layer 2
Layer 3
log(1 + weight-token count)
Exact rule-set signature
d
Single-task tokens
Relative accuracy
Retained rule set
Rule 1
Rule 2
Rule 3
Rule 4
Rule 5
e
Two-task tokens
Relative accuracy
Retained rule set
f
Multi-task tokens (3–5 tasks)
Relative accuracy
Retained rule set